

The Setup A Convention

What the “Setup A convention of Goudy and coworkers” means when it is cited for the protonic Fock operators of a multicomponent Hartree–Fock reference with several quantum protons, what it includes and excludes, why the choice matters for triples corrections and derivatives, and how to tell which setup a code uses.

Written for a graduate student who has read the NEO coefficients and NEO gradient notes

Protons treated as distinguishable particles in a Hartree product

NOTATION

α, β	labels for the quantum nuclei; $p\alpha$ the orbital space of nucleus α .
$\hat{h}^{p\alpha}$	core Hamiltonian of nucleus α : its kinetic energy and its Coulomb interaction with the classical nuclei.
$\hat{v}_{\text{HF}}^{\text{ep}, p\alpha}$	the mean field of the electrons acting on nucleus α , attractive.
$\hat{v}_{\text{HF}}^{p\beta p\alpha}$	the Coulomb mean field of nucleus β acting on nucleus α , repulsive.
$\hat{J}, \hat{K}$	Coulomb and exchange operators, in the sense of ordinary Hartree–Fock theory.

THE WHOLE ANSWER IN THREE LINES

- 1 Setup A defines one Fock operator per quantum nucleus: its core Hamiltonian, the electrons' mean field, and the Coulomb mean field of the *other* quantum nuclei. No proton–proton exchange and no self-interaction.
- 2 Every convention of this kind gives the same NEO-HF energy, because the terms they differ by cancel exactly in the energy. They differ in the virtual protonic orbitals and their energies.
- 3 That is why the choice must be stated: perturbative triples, CC3-type triples and every orbital-energy denominator depend on it, and a gradient Lagrangian inherits it through the Fock operators, the orbital Hessian and the effective densities.

1 Where the label appears

The analytic gradient paper for multicomponent CCSD by Tucker and Brorsen defines its reference in Section 2.1 with the sentence

“The reference is the multicomponent HF wave function with Fock operators defined using the Setup A convention of Goudy and coworkers. In this convention, the Fock operator for quantum nucleus α is”

KEY RESULT · THE SETUP A PROTONIC FOCK OPERATOR (EQ. 3 OF THE GRADIENT PAPER)

$$\hat{F}^{p\alpha} = \hat{h}^{p\alpha} + \hat{v}_{\text{HF}}^{\text{ep},p\alpha} + \sum_{\beta \neq \alpha} \hat{v}_{\text{HF}}^{p\beta p\alpha}. \quad (1)$$

The citation is to Goudy, Pavošević and Hammes-Schiffer, *J. Chem. Phys.* **163**, 224119 (2025), the paper that introduced triple excitations into NEO coupled cluster theory for systems with several quantum protons. The gradient paper also states the setting the convention belongs to: the quantum nuclei are distinguishable, the reference is the product

$$|0\rangle = |0^e\rangle \otimes \prod_{\alpha} |0^{p\alpha}\rangle, \quad (2)$$

with $|0^{p\alpha}\rangle$ the single occupied orbital of nucleus α , each nucleus has its own orbital space, and the proton–proton part of the two-particle density has no exchange term.

What the Goudy paper says in its main text

The setups are not defined in the main text of the Goudy paper; they live in its supplementary material. The main text does say what the distinction is about, in the results section:

“The NEO-CCSD($T_{\text{eep},\text{epp}}$) energies were found to be sensitive to the inclusion or exclusion of the proton–proton interaction term in the Hamiltonian for systems containing only a single quantum proton. Although the contribution of this term to the NEO-HF energy is zero for systems with only a single quantum proton, its presence influences the virtual orbitals and the perturbative energy correction. This behavior has been observed previously for multicomponent perturbative methods. For the proton affinity calculations in the main text, the proton–proton interaction was included when at least one of the two energy calculations (A or AH^+) involved multiple quantum protons but otherwise was not included. More information on the effect of including or excluding the proton–proton interaction term in the Hamiltonian and on its connection to the definitions of the zeroth-order Hamiltonian and perturbation is provided in the supplementary material.”

The supplementary material is described as containing an “analysis of the inclusion or exclusion of the proton–proton interaction term in the Hamiltonian and its connection to the definitions of the zeroth-order Hamiltonian and perturbation”. The data deposited with the paper on Zenodo labels every single-proton calculation VPP1 or VPP0, which its README explains as “whether or not proton-proton interaction was included for the calculations involving a single quantum proton”, adding that “for multiple quantum proton calculations, proton-proton interaction is always included”; the Q-Chem inputs carry the corresponding keyword, and one output warns that the printed electron–proton MP2 energy is not meaningful for a single proton when the term is switched on. The deposited LaTeX for the NEO-CC energy also matches, term for term and coefficient for coefficient, the expression derived in the NEO coefficients notes of this series.

2 What the convention contains

Read (1) term by term.

- **One Fock operator per nucleus.** Nucleus α has its own orbital set, its own occupied orbital and its own virtuals, and its own eigenvalue problem. There is no shared protonic Fock operator.

- **The core Hamiltonian** carries the nucleus's kinetic energy, with its own mass, and its Coulomb repulsion from the classical nuclei. It is the only place the classical nuclei act on the quantum proton directly.
- **The electrons enter as a mean field**, the attractive potential of the electronic Hartree–Fock density, which couples the two species at the mean-field level and is the origin of the off-diagonal blocks of the NEO-HF orbital Hessian.
- **The other quantum nuclei enter as Coulomb mean fields only.** The sum runs over $\beta \neq \alpha$, so the nucleus does not see its own density, and there is no exchange operator, because distinguishable particles have none.

Two things are therefore absent that an unrestricted high-spin Hartree–Fock treatment of identical protons would contain: a self-Coulomb term $\hat{J}_\alpha$ acting on nucleus α , and an exchange term $-\hat{K}_\beta$ from every nucleus, including $\beta = \alpha$. Section 4 returns to these.

3 As implemented in a production code

The NEO coupled cluster package these notes accompany, which builds on the NEO extension of PySCF, implements (1) literally. Each quantum nucleus is a separate SCF component, documented as a distinguishable single particle that does not see a same-species two-particle potential: its own Coulomb and exchange contributions are returned as zero. The interactions between components, electron–nucleus and nucleus–nucleus alike, are Coulomb potentials computed from the other component's density and added to the component's Fock matrix. The second-order SCF solver sets the same-nucleus orbital-Hessian block to zero for this reason, and the CPHF module's protonic orbital Hessian has Coulomb coupling only between distinct nuclei, with no same-nucleus two-particle integrals. The multi-proton CC3 and CCSDT solvers in the same package are built on this reference and use its canonical orbital energies in their triples denominators, which is where the convention enters the correlated methods.

4 What the two setups are about

CAUTION · PARTLY VERIFIED

The main text of the Goudy paper and its deposited data were read for these notes; the supplementary material, where the setups are defined and lettered, was not available. What the distinction concerns is therefore established; which of the two conventions carries the letter A, and whether the setups are stated as alternative Fock operators or as alternative partitionings of the Hamiltonian with the same Fock operator, is not.

From the passage quoted in Section 1, the two setups differ in whether the proton–proton interaction term is part of the Fock operator, and so of the zeroth-order Hamiltonian that defines the orbital energies, or is left out of it. What that term is depends on how many quantum protons there are.

- **One quantum proton.** In a determinant-based formulation with high-spin protons, the proton–proton term is the proton’s interaction with its own density, Coulomb minus exchange. For the one occupied orbital the two pieces cancel, so the NEO-HF energy is unchanged, but the operator is not zero on the virtual space, so the virtual orbitals and their energies change. This is the case the Goudy paper describes, and the one its VPP0 and VPP1 labels switch between.
- **Several quantum protons.** The term also contains the interaction between distinct protons, which does contribute to the energy and is always included. Equation (1) of the gradient paper keeps exactly this part, the Coulomb mean field of the other nuclei, and has no self-term, which is the Hartree-product form of the operator.

Independent support for the same picture comes from Chow, Paenurk and Hammes-Schiffer, *J. Chem. Phys.* **164**, 074103 (2026), who compare the Slater-determinant representation of several quantum protons, with one shared protonic Fock operator carrying an unrestricted Coulomb sum and an exchange operator including the self-terms, against the Hartree-product representation, with one Fock operator per nucleus, no exchange, and the Coulomb sum restricted to the other nuclei. They note that the self-Coulomb and self-exchange terms cancel exactly in the energy and can be added back without changing it, and that both representations give the same NEO-SCF energy.

On the evidence available, Setup A as used in the gradient paper is the Hartree-product convention of (1): inter-proton Coulomb mean field in the Fock operator, no exchange, no self-interaction. The other setup is the one in which the single-proton self-term is retained in the Fock operator, or equivalently in which the proton–proton interaction sits on the zeroth-order side of the partitioning. That assignment of the letters should be confirmed against the supplementary material before it is relied on.

5 Why the choice matters

All of these conventions give the same NEO-HF energy and the same occupied protonic orbitals, because for a single particle in an orbital the Coulomb and exchange self-terms are equal and cancel, and for distinct protons with non-overlapping orbitals the exchange is negligible. What they do not share is the virtual space: adding $\hat{J}_\alpha - \hat{K}_\alpha$ to the Fock operator of nucleus α leaves its occupied orbital an eigenfunction with the same eigenvalue but changes every virtual orbital and every virtual orbital energy. The Goudy paper states this directly for its perturbative triples and notes that the same sensitivity was seen earlier for multicomponent perturbation theory,

citing Swalina, Pak and Hammes-Schiffer (2005). Consequences follow for everything built on the virtuals.

QUANTITY	DEPENDS ON THE SETUP	REASON
NEO-HF energy	no	the self-terms cancel in the energy
Occupied protonic orbitals	no	they remain eigenfunctions with the same eigenvalue
Virtual protonic orbitals and energies	yes	the added operator is not zero on the virtual space
NEO-MP2, NEO-CCSD(T), CC3-type triples	yes	orbital-energy denominators enter explicitly
Converged NEO-CCSD energy	no, if the full Fock matrix is used	CCSD is invariant to rotations within the protonic virtual space and the off-diagonal Fock terms are kept
NEO-CCSD convergence rate	yes	the denominators of the fixed-point iteration change
Gradient Lagrangian pieces	yes	the Fock operators, their derivatives, the orbital Hessian blocks and the effective densities all carry the convention

The last row is why a gradient paper has to name the convention even when its correlated energy does not depend on it. The Lagrangian of the NEO gradient notes attaches the Brillouin conditions $f_{AI}^{p\alpha} = 0$ of each nucleus, differentiates each Fock matrix with respect to the perturbation, and builds the protonic orbital Hessian blocks; in Setup A the same-nucleus block

contains only orbital-energy differences and the coupling between nuclei is Coulomb only, and the effective proton–proton density carries no exchange. With the self-terms present, every one of these objects would have additional pieces, and a gradient derived in one convention would be wrong in the other.

6 How to tell which setup a code uses

- **Add the self-terms and watch the energy.** Including $\hat{J}_\alpha - \hat{K}_\alpha$ in the Fock operator of nucleus α must leave the NEO-HF energy unchanged to the convergence threshold. If it does not, the code is not computing a Hartree–Fock energy at all.
 - **Watch the virtual orbital energies.** They will shift. A code in Setup A reports the unshifted values; the shift is the size of the self-interaction on the virtual space and is not small for compact protonic basis sets.
 - **Run NEO-CCSD and NEO-CCSD(T) both ways.** The CCSD energy should agree to the amplitude convergence threshold, provided the implementation keeps the full Fock matrix; the (T) correction will differ. A CCSD difference points to a code that assumes a diagonal protonic Fock matrix.
 - **Check the CPHF matrix.** In Setup A the protonic diagonal block for a single quantum nucleus reduces to orbital-energy differences, and the proton–proton coupling block is pure Coulomb.
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7 Sources and what was verified

The quotation and Eq. (1) in Section 1 are taken from the open-access text of the gradient paper, and the implementation statements in Section 3 from the production code these notes accompany. The main text of the Goudy paper was read in full, and its deposited data on Zenodo were downloaded and inspected: the README of the proton-affinity data, the Q-Chem keyword in the inputs and outputs, and the LaTeX for the energy and amplitude equations. Its supplementary material, which defines and letters the setups, was not available, so the assignment of the letters in Section 4 remains an inference.

1. G. B. Tucker and K. R. Brorsen, “An Analytic Nuclear Energy Gradient for Multicomponent CCSD”, *J. Chem. Theory Comput.* **22**, 8813 (2026). Section 2.1, Eqs. (2) and (3), and reference 31.
2. R. J. Goudy, F. Pavošević and S. Hammes-Schiffer, “Triple excitations in nuclear–electronic orbital coupled cluster theory for multiple quantum protons”, *J. Chem. Phys.* **163**, 224119 (2025). The source of the Setup A label; the main text quoted in Section 1.

3. R. J. Goudy, F. Pavošević and S. Hammes-Schiffer, data for the above, Zenodo, doi:10.5281/zenodo.17781752 (2025), CC BY 4.0. Inputs, outputs, the VPP0/VPP1 labels and the equation files.
4. M. Chow, E. Paenurk and S. Hammes-Schiffer, “Efficient and robust optimization of nuclear and electronic orbitals within the nuclear Hartree product representation”, *J. Chem. Phys.* **164**, 074103 (2026). The Slater-determinant and Hartree-product Fock operators and the cancellation of the self-terms.
5. C. Swalina, M. V. Pak and S. Hammes-Schiffer, “Alternative formulation of many-body perturbation theory for electron–proton correlation”, *Chem. Phys. Lett.* **404**, 394 (2005). The earlier observation that the partitioning affects multicomponent perturbative energies.
6. F. Pavošević, T. Culpitt and S. Hammes-Schiffer, *Chem. Rev.* **120**, 4222 (2020). The NEO framework and NEO-HF.

Equation (1) is quoted from the gradient paper with its notation, and the block quotation in Section 1 from the main text of the Goudy paper. The assignment of the letters A and B in Section 4 is an inference and should be replaced by the definitions in the Goudy supplementary material when it is at hand.